



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/LMI/437340

Date:

29/08/2023(16:30:32)

Subject:Revision of LMI/OGN/OPS/ELEC/021, MAINTENANCE AND TESTING OF STATION EARTHING SYSTEMS

Revision of LMI/OGN/OPS/ELEC/021

MAINTENANCE AND TESTING OF STATION EARTHING SYSTEMS

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

OGN/OPS/ELEC/021 has been revised by COS in July 2023. No changes envisaged by COS.

LMI is reviewed. Changes have been made in clause 2.0 "SCOPE", changes are highlighted in blue colour.

Put up for kind approval.

Attachments:

1. LMI/OGN/OPS/ELEC/021, Rev 7, Aug 2023
2. OGN/OPS/ELEC/021, Rev 1, June 2021

Submitted by:

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On:29/08/2023(16:30:32) **Ref:**SMTTP/ELECT MAINT/2023-24/LMI/437340

Note No:2

revised LMI is in order.

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Note No:3

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LMI revised as discussed.

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Note No:5

To be reviewed by all sections

LMI found in order as per OGN/21 Revised June 2021

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Note No:6

Please include in the scope of building earthing as:

1. CHP control room,
2. Coal bhavan,
3. Fuel bhavan,
4. Crusher house#1 &2,
5. Track hopper#1&2 &
6. Track hopper control room

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Note No:7

LMI will be modified after routing to all the sections.

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Note No:10

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Note No:12

Reviewed the LMI w.r.t OGN/21 revised June 2021.

1. CHP control room
2. Coal bhavan
3. Fuel bhavan
4. Crusher house#1 &2
5. Track hopper#1&2
6. Track hopper control room

All 6 Above locations may be included in the scope of LMI.

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Note No:13

the mentioned areas are being covered in the LMI.

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Note No:14

LMI reviewed and found in order

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Note No:18

May please approve.

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Note No:19

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NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/ELEC/021
ISSUE NO: 1, REVISION: 7
DATE: AUG 2023

**MAINTENANCE AND
TESTING OF
STATION EARTHING SYSTEMS**

APPROVED FOR

IMPLEMENTATION BY -----

Group General Manager (Simhadri)

NTPC Limited
SIMHADRI SUPER THERMAL POWER STATION
OPERATION & MAINTENANCE DEPARTMENT

LMI (Name) :	MAINTENANCE AND TESTING OF STATION EARTHING SYSTEMS		
LMI No.	: LMI/OGN/OPS/ELEC/021		
Issue No.	1	Revision : 7	Date: AUG 2023

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MAINTENANCE AND TESTING OF STATION EARTHING SYSTEMS

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1.0 **OBJECTIVE :-**

The main objective of an earthing system is to provide as nearly as possible a surface under and around a station which shall be at uniform potential and as nearly zero or absolute earth potential as possible. By providing the earth surface of uniform potential under and surrounding the station, there can exist no difference of potential in a short distance big enough to shock or injure an attendant when short circuits or other abnormal occurrences take place.

Earthing associated with current carrying conductors is normally essential to the security of the system and is generally known as system earthing while earthing of non –current carrying metal works and conductors is essential to the safety of human life , animals and property, and is generally known as equipment earthing .

It therefore, follows that for the safety of the plant and personnel, it is essential that the earthing system at each Power Station should remain effective throughout the life of the station. It is difficult in many cases to make a check of continuity after installation. The system, therefore, has to be robust and protected from mechanical damage and corrosion, where necessary. Any joints should be capable of retaining low resistance after many passages of fault current.

The Main Earthing System (MES) has two main functions: -

- i) To safely discharge earth system fault currents to the ground in all conditions, without change in resistance.
- ii) To provide a sufficiently low impedance path into the ground for the designed value of primary system earth fault current, so as to limit the resultant rise of earth potential to an acceptable level. The possible rise in potential between the station earth system and the general body of earth, should be as low as economically possible.

The general structural steelwork of the Power Station can be classed as an adventitious earth mat. The difficulties of adequately testing and monitoring the structure for use as a suitable earth electrode rules out relying on this as an alternative to the MES. Earthing and bonding by nature, does not automatically require maintenance although this may be necessary to ensure it functions effectively at all times. During most of its life it is quiescent, and attention is not drawn to incipient faults, deterioration or the dangers resulting from neglect. When called upon to serve its function during electrical faults, the failure of the earth system can result in damage of catastrophic proportions.

It follows therefore, that planned routine maintenance, detailed inspection and recording of results can ensure the safe operation of the earthing system at a Power Station site. In multiple earthed systems, as is

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commonly found in Power Stations, it is not easy to measure the resistance of all individual components of the buried and above ground earth system, unless provision was made at the time of installation. The whole installation must therefore be robust and protected against corrosion and mechanical damage.

Our installations have not presented serious problems and have shown that initial design and installation standards have been satisfactory. However, a system of routine electrical testing and some visual inspection is necessary to achieve continued reliable performance.

The objective of this LMI is to ensure the healthiness of the station earthing system at NTPC Simhadri

2.0 SCOPE :

The document covers the testing and maintenance of station main earthing systems. It covers the system earthing, equipment earthing.

Equipment earthing and system earthing in the following areas are covered

- 400 KV switch yard of stage-1&2.
- All HT /LT Transformers in the main plant and off sites of stage-1&2.
- 33KV sweet water, sea water, raw water, ash water recirculation Switchgears and switchyards of stage-1&2.

Building earthing for lightning protection in the following areas is covered.

- CW pump house of stage-1&2.
- CPU regeneration plant and DM make up pump house of stage-1&2.
- DM plant, PT plant and Chlorination plant buildings of stage-1&2.
- Fire water tanks and Fire water pump house of stage-1&2.
- Ash water make up, Ash handling control building and Vacuum pumps switchgear room of stage-1&2.
- TG building of stage-1&2.
- Unit 1 , Unit 2 ,Unit 3 & Unit 4 Boilers
- Cooling towers 1, 2,3&4 , Chimneys of stage-1&2.
- Fuel oil pump house building , HFO tanks and LDO tanks of stage-1&2.
- Effluent disposal pump house of stage-1&2.
- Sankalp Building, Sankalp annexe building
- CHP control room ,Track hopper & Track hopper control room of stage-1&2.

- Ash silos [switchgear of stage-1&2](#)
- Hydrogen plant
- Admin building, samanvaya annexe building, IT building, FES building

3.0 REFERENCE DOCUMENT :

Operation Guidance Note OGN/OPS/ELEC/021.

REV: 1 Date : June 2021

4.0 SUPERSEDED DOCUMENTS :

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5.0 SPECIAL INSTRUCTIONS TO BE FOLLOWED

[Annual measurement of earth resistance should be carried out but local circumstances, based on experience, may justify an increase or decrease in this interval which should not be of a greater frequency than once in 2 years.](#)

Instruction	Responsibility	Document Ref
Measurement of Earth resistance	Respective Sectional Heads of EM	LMI/E-021

5.1 INSPECTION :

[An earth survey should ensure compliance with the main objectives of IS:3043-2018, Indian Standard Code of Practice for Earthing, and should be supplementary to the Power Station maintenance programme on above ground earthing.](#)

[The accessibility to earthing electrodes and the associated connections varies from station to station and also within stations. Electrode pits often have been buried, built over and in some instances not installed as the drawing records indicate. It is important that accurate records and drawings are kept of all electrode pit and link locations. The location of buried or built over pits is often undeterminable, the only indication to their existence being conductor from the electrodes to the earthing system as it enters the station buildings at basement, sub-basement or cable tunnel level.](#)

Accessible electrode pits should be examined for the condition of the conductor connections to the electrodes, corrosion of fixings/fastenings, and the general condition of the pit civil work and covers. The electrodes types will either be groups of rods, groups of cast iron pipes or sheet steel pile or group of piles. The most common form of deterioration takes place at the joint between the conductor and the steel pile or cast iron pipe, where corrosion increases the joint resistance. These joints should be protected by the application of a proprietary anti-corrosion system.

5.2 INDIVIDUAL ELECTRODES :

Most electrode positions have isolating links in the associated interconnecting conductor (s) to facilitate testing. Electrodes that have been buried or built over should be tested from the point of connection where the conductor from the electrode joins the MES, if this point is accessible. Most station earthing system conductors have radial connections to the external electrodes. Priority in testing should be given to these types of electrode, this being the only way to determine their condition.

The individual electrodes should be tested by comparison. The test procedure should use a Null Balance Earth Resistance “Megger” and should be carried out in accordance with Indian Standards Code of Practice for Earthing IS:3043-2018. Individual electrode resistances should be as low as possible preferable around 0.5 ohms.

When testing, only one group of rods or a plate should be isolated at time, such that remaining electrodes are capable of discharging the prevailing earth fault current.

Disconnection and testing should not be undertaken at times of lightning risk. Prior to disconnecting an electrode, it is imperative that a check is carried out to confirm that the system terminating at the point is fully interconnected to other adjacent earth electrodes via the MES. The earth current flowing in the termination should be measured.

If any doubt about connection to the MES exists, the MES side of the termination point or link should be connected to the MES with the temporary conductors of a cross section equivalent to the termination conductor. Additionally, before removing the termination from the electrode, the test link or disconnection point should be bridged with suitable value resistor. With the link removed, the voltage across the resistor should be measured. Any voltage across the resistor would indicate a true power flow with the existence of a neutral earthed transformer system connected in isolation from the MES and the earth electrode termination point being investigated. After the disconnection has been made, the bridging resistor can be removed from the termination, leaving the electrode isolated and ready for test.

Following tests, the bolted link connections should be cleaned, remade and checked for continuity. Readings below 10 micro ohms are satisfactory.

5.3 ROUTINE TESTING AND INSPECTION OF ABOVE GROUND EARTHING :

Above ground earthing is subject to mechanical damage, theft and manual interference, thus all the following items should be visually inspected on an annual basis: -

- a) Earth link connections between buried electrodes and the rest of the MES.
- b) Earth link connections between adjacent power station and sub-station.
- c) Connections at buried earth electrode to the adjacent MES.
- d) Link connections between various sections of the MES.
- e) Main earthing conductor runs.
- f) Transformer neutral connections to the MES.
- g) Generator earthing systems including associated earthing resistors/transformers and the connection of the latter items to the MES.
- h) The earth terminal to sheath connections of power cables paying particular attention to single core cables of 3.3 KV and above.
- i) The connections between switchboard earth conductors, generator casings, phase isolated busbar casings, steel support structures for outdoor high voltage overhead conductors, station structural steelwork and the MES.
- j) Battery system positive pole connections to the MES.
- k) Connections from lightning protection to the MES.
- l) Connections between fuel oil storage tanks and driven rod electrodes.
- m) Earthing connections and systems associated with flammable gas storage compounds.
- n) Earthing of motors, switchgear etc.
- o) Earthing of switchyard equipment such as isolator earth switch lighting arrestor CVT NGR & Shunt reactors etc.
- p) Earthing of Switchyard fencing, railway tracks, cable racks/trays & pipe racks.

In addition to these visual inspections, electrical resistance measurements should be carried out at annual intervals on items (a), (b) and (c). An ohm meter ("Ductor" type) with as high a testing current as possible (10 Amps), should be used, and all results recorded.

6.0 METHODOLOGY OF IMPLEMENTATION :

- Identification of the earthing electrodes connected directly or through test links to the earth mat.
- Inspection of the earthing electrode for any corrosion and application of anti-corrosion paint wherever necessary.
- Measure the earth resistance of equipment earthing or system earthing either at the point of connection to the earth grid or at an approachable point.
- If the earthing is through bolted test links, open the test link during shut down of the equipment. Check the healthiness of the link.
- Measure the earth resistance of the pipe or rod electrodes and earthing grid after disconnecting the links.
- Clean and then re connect the link. Measure the earth resistance. *Individual electrode resistances should be as low as possible preferable around 0.5 ohms.*
- Measure the contact resistance of the test link. The contact resistance should be preferably less than 10 micro ohms or it should be compared to the commissioning / previous values.
- For easy reference a drawing showing the equipment earthing through a rod electrode type earth pit is enclosed.
- For earthing used for lightning protection of buildings and earthing of bus ducts. Visual inspection of earthing electrodes / strips is to be done. Check for any corrosion and apply anti-corrosion paint wherever necessary. Measure the earth resistance and record the values.
- For all the electrodes which have got more than specified value, necessary corrective action should be taken by the concerned area incharge and after correction, values are to be measured and should be within the allowable limits.

6.1 **RESULT OF TESTS :**

Electrode resistance depends on various factors, such as electrode dimension, soil type and ground conditions etc. If all the parameters of the installation and ground type are known, it is possible to make an estimation of the electrode resistance and the overall site resistance. Reliance on absolute values should not be made; changes in value should be noted and judgement made on trends, modified by local conditions such as prolonged dry weather. Testing earthing system should be carried out during dry season to get conservative results. Values for electrode resistance measured before commissioning and subsequent results should be available. Seasonal variations should not account for more than a 25% change in results. If variations are greater than this then consideration should be given to excavating or replacing the electrode in question. In certain instances, especially where annual rainfall has for consecutive years been low resulting in a lowering of the water table, electrode condition may be recovered by salt treatment of the soil adjacent to the electrode. Guidance for this is given in Indian Standards Code of Practice for Earthing, IS:3043-2018.

Comparison of joints tested for resistance can be made using a similar conductor free from joints and measuring over the same test length.

7.0 **EARTHING ADEQUACY STUDY:**

Stations completing 20 years of service and where new units are being added, which have not been envisaged initially at the time of first stage or subsequent stage engineering, are required to take up earthing adequacy study and based on result necessary action to be planned.

Earthing Adequacy study was done by M/s Manav energy private limited at NTPC

Simhadri and submitted its report in March 2021.

8.0 **RESPONSIBILITY CENTRE FOR IMPLEMENTATION :**

Area engineer of concerned area is responsible for implementation and overall responsibility lies with the sectional head.

9.0 **EXCEPTION REPORTING AND DISPENSATION OF DEVIATIONS**

Any exceptions / deviations shall be reported to HOD (EM).

10.0 **RECORD KEEPING :**

Records will be maintained by concerned area maintenance groups in respective files with file names LMI/E-021/1802, LMI/E-021 / 1803, LMI/E-021/1804, LMI/E-021/1805, LMI/E-021/1806

11.0 REVIEW:

This LMI document will be reviewed once in every two years by Head of the Project, NTPC Simhadri.

12.0 RECORDS REVISION:

1. Clause 1.0 Objective
2. Clause 2.0 Scope
3. Clause 5.0 Special instructions to be followed
4. Added Clause 5.1 Inspection
5. Added Clause 5.2 Individual electrodes
6. Added Clause 5.3 Routine testing and inspection of above ground earthing.
7. Added Clause 6.1 Result of tests.

Changes are highlighted in blue color

12.0 APPENDIX**SAMPLE CHECKLIST FORMAT OF VARIOUS BUILDINGS****Recommended:** Earth Resistance value should be *around 0.5 ohm**Contact resistance of test link should be less than 10 micro ohms*

Areas : Sankalp Building, Chimney, FOPH, Cooling Towers, TG & Boiler.

Measured Date	Pit /Electrode No.	Measured Resistance in Ω	REMARKS	LOCATION: MP & SANKALP BLD.
	1			
	2			
	3			
	4			

Contact resistance of test link:

Measured Date	Pit /Electrode No.	Measured Resistance in Ω	REMARKS	LOCATION: FOPH
	1			
	2			
	3			
	4			
	5			
	6			

Contact resistance of test link:

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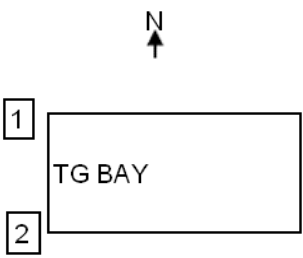
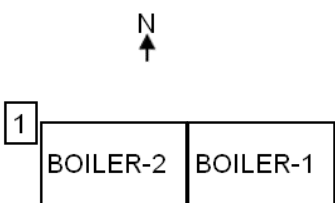
Measured Date	Pit /Electrode No.	Measured Resistance in Ω	REMARKS	LOCATION : COOLING TOWERS
	1			
	2			
	3			
	4			
	5			
	6			
	1			
	2			
	3			
	4			
	5			

Contact resistance of test link:

Measured Date	Pit /Electrode No.	Measured Resistance in Ω	REMARKS	LOCATION : CHIMNEY
	1			
	2			
	3			
	4			

Contact resistance of test link:

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Measured Date	Pit /Electrode No.	Measured Resistance in Ω	REMARKS	LOCATION : TG & BOILER
	1 2			 A diagram showing a rectangular area labeled 'TG BAY'. Above the rectangle is an arrow pointing upwards with the letter 'N' above it. To the left of the rectangle, there are two small boxes containing the numbers '1' and '2' stacked vertically.
	1			 A diagram showing two adjacent rectangular areas labeled 'BOILER-2' and 'BOILER-1'. Above the rectangles is an arrow pointing upwards with the letter 'N' above it. To the left of the 'BOILER-2' rectangle, there is a small box containing the number '1'.

Contact resistance of test link: